

ABY SAJI PAPPACHAN

TECHNICAL LEAD

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Right to Work in the UK, no sponsorship required | Available immediately

SUMMARY

Technical Lead with 8 years across four startups, promoted in 2025 after two consecutive years rated top performer, leading a team of 4 and selected for Architect before relocating to the UK. Two things have defined my career. I am the person people bring "could we actually do this?" to, from sales and product as often as from engineering, and I answer it with a working prototype rather than an opinion. And I get handed the messy shared problem: the framework, the standard or the platform that makes something stop being hard for everyone else. A Zero Trust product taken from nothing to 10+ enterprise customers, a browser extension serving 2 million users, and 300+ third-party integrations where the difficulty was always the aggregate. TypeScript, Node.js, Go, Python and React on AWS. Every company I have worked for has asked me to come back.

CORE SKILLS

Leadership	Team leadership and mentoring, design and architecture review, engineering standards, technical interviewing, cross-team delivery, HLD and LLD, technical feasibility and proof of concept
Languages	TypeScript, JavaScript, Node.js, Go, Python, Java, SQL, Lua, Rust (projects)
Backend	NestJS, Spring Boot (incl. WebFlux), FastAPI, REST, GraphQL, microservices, event-driven architecture, framework and SDK design, system design
Web & Client	React, Next.js, Angular, Electron, Tauri, cross-browser extensions (Chromium and Firefox, MV2 and MV3)
Cloud & Infra	AWS (ECS, SQS, SNS, S3), Docker, Kubernetes, CI/CD, release management, performance and reliability testing
Data & Search	PostgreSQL, MySQL, MongoDB, OpenSearch / Elasticsearch, Redis, graph databases, Apache Airflow, Airbyte, DuckDB, ETL, data modelling, query optimisation
AI & LLM	MCP (Model Context Protocol), LLMs, RAG, LangChain, AI agents, semantic search, embeddings
Security	Zero Trust Network Access, WireGuard, SOCKS5, HAProxy, Semgrep, Gitleaks, Trivy, SAST and secret scanning

EXPERIENCE

Technical Lead | Josys Digital Technologies | Bengaluru, India | Aug 2022 to Jun 2026

Senior Software Engineer Aug 2022 to Mar 2025. Promoted with a raise three months after joining, to Technical Lead in 2025, and selected for Architect in 2026. Top performer rating two consecutive years. Left to relocate to the UK.

Leadership and Engineering Standards

- **Led and mentored a team of 4 engineers** through design reviews, pairing and code review, and became the go-to reviewer for architecture decisions outside my own project scope.
- **Given ownership of code review and engineering standards across every project in the company** after diagnosing a production memory leak in another team's service that nobody else could find. Also ran internal sessions on Node.js failure modes and error handling.
- **Coordinated delivery across 3 product teams** as backend architecture owner, resolving design disputes and setting the interface contracts between services.
- **Ran technical interviews and onboarding** for new backend hires, and set the review standards the team still works to.
- **Cut new-engineer onboarding from 6 weeks to 3** by winning the case for NestJS as the company-wide backend standard across 12+ microservices, which meant persuading several teams rather than mandating it. It remains the standard.
- **Became the company's first stop for feasibility.** Architects, product managers and sales brought ideas for assessment; my proof of concept and documentation repeatedly became the base implementation months later, including the OpenSearch index schema adopted unchanged when the capability was productised.

Platforms and Production

- **Stopped repeated production crashes within minutes** by tracing them to a node-fetch memory leak on aborted requests, then shipping a drop-in streaming wrapper applied by find and replace, avoiding an unupgradable dependency change across 300+ integration sources.
- **Cut testing infrastructure cost 35% and removed the need for 300+ live SaaS test accounts** by building a simulation framework in NestJS reproducing every integrated provider, including rate limits, pagination quirks and delays. It never required a bug fix across its lifetime and became the customer-facing production sandbox.
- **Consolidated 8 fragmented notification flows into one service at 50,000 messages a day**, built solo on REST and AWS SNS into SQS across email, Slack and Teams, split into on-demand and delayed batch paths. Templating is convention-driven, so adding a template is dropping in a file. Later rewrote the two highest-volume channels in Go for throughput.
- **Removed roughly 70% of hand-written data-access code** by building data-gateway, a GraphQL endpoint spanning two MySQL databases, OpenSearch and MongoDB that **generates its schema from table specifications at build time**, plus a runtime GraphQL to raw SQL compiler adopted directly by other microservices.
- **Protected 300+ third-party APIs from being overwhelmed** with a rate limiting framework configured entirely through database entries, correct **across horizontally scaled instances** rather than per-process, enforceable at global, container, service-type or individual app level.
- **Scaled a cross-browser extension to over 2 million users** from a single codebase across Chromium, Firefox and Manifest V2 and V3, adding **silent activation** so administrators force-push it with keys and it authorises itself, plus the PowerShell, macOS and Linux deployment scripts.
- **Led the security project and team**, building a Go service orchestrating Semgrep, Gitleaks and Trivy across the GitHub repository estate with scan scheduling, concurrent execution, result normalisation across three formats, and a dashboard API by repository and severity.
- **Unblocked every UI team from backend engineers** with a Node.js proxy forwarding local frontend work to deployed environments with cookies and auth intact, plus a companion database proxy for tunnelled access to MySQL, Postgres, MongoDB, Redis and OpenSearch.

Data, Analytics and AI

- **Improved query performance roughly 70% on datasets above 50M rows and cut analytics spend 45%** by rebuilding the reporting platform on **Airbyte, DuckDB and Apache Airflow**, and took the most-used business reports from **25 minutes to under 2** through data modelling changes that made the expensive query unnecessary. Owned ETL design across **15+ data sources**.
- **Cut integration time for a new service from 2 weeks to 2 days** by building MCP clients and servers standardising how AI agents reach backend services.
- **Saved the engineering team 8 hours a week, a 60% reduction in manual querying**, by shipping the company's first LLM assistant in **Python and LangChain** for context-aware retrieval across production data sources.
- **Enabled semantic queries the relational schema could not support** by introducing a graph database layer modelling relationships across 20+ entity types.

Software Engineer (MTS-1) | InstaSafe | Bengaluru, India | May 2019 to Aug 2022

Hired directly onto the Zero Trust Network Access product, working under the lead architect designing it.

- **Contributed a new enterprise product line that helped onboard 10+ enterprise customers** by designing and building a **Zero Trust Network Access solution from scratch**, interface through to the dashboard across backend and frontend, including a widget layout whose size and ordering are backend-controlled and persisted.
- **Led 3 engineers delivering 6 security product enhancements**, owning each from design through production.
- **Cut enterprise onboarding time 40%** by researching and building an **agentless gateway** giving browser access to private web applications with no desktop client, using Node.js, TypeScript and Lua scripting over HAProxy to block traffic classes, redirect and force logout. Still in production.
- **Owned the cross-platform Electron desktop agent** used by roughly **5,000 users** on macOS, Windows and Linux. Each webview runs behind **its own SOCKS5 proxy**, created locally and injected after the agent verifies approval, access rights and gateway location, so every application routes through the nearest gateway. Backend messaging over **AWS SQS**. Also carried SMB file browsing with in-place editing, clipboard blocking, screen-capture blocking and an in-app remote Windows session.
- **Built a cross-platform VPN service in Go** running as a privileged system service on macOS, Windows and Linux over **WireGuard**, installed by the agent for the correct OS and architecture and driven over IPC for **full or split tunnels** across multiple applications and gateways from one configuration.
- **Led the migration of the backend microservice estate from Java (Spring Boot WebFlux) to Go**, rewriting services one at a time so both stacks ran side by side rather than forcing a single high-risk cutover.

Software Engineer | GrowthEnabler | Bengaluru, India | Aug 2018 to May 2019

The only junior engineer on a team where colleagues had between 8 and 30 years of experience.

- Hired as a **frontend engineer on AngularJS** and shipped a fix on the first day, then moved into backend work on **Java, Spring Boot and MySQL** within three months.
- Took ownership of **PDF report generation** end to end, frontend rendering and backend generation using **JHipster**, and delivered the in-product tour guide across frontend and backend. This is where I learned the engineering standards I still work to.

Software Engineer | Haappy | Bengaluru, India | Jan 2018 to Aug 2018

First role. Joined a live-streaming product company the week its entire engineering team departed, inheriting a Java and Angular codebase with nobody left who could run it.

- Learned **Angular** from scratch and delivered a news-channel website end to end, then became the engineer the business routed unfamiliar work to, delivering end-to-end projects in **C++, R, Python and C#**. Also built an **Ionic** mobile app and a CRM in **Node.js, Angular and MongoDB**.

OPEN SOURCE

- **AgentKit** (abyabyss.github.io/agentkit): MIT-licensed **config-first headless agent runtime for TypeScript**, published as separate packages for core, server, client, React and workspace. Multi-provider with mid-session model hot-swapping, full MCP support over stdio and HTTP with Zod-typed tools, human-in-the-loop plan mode and approval gates, and a **record and replay harness** so agent behaviour is tested deterministically in CI with no API keys and no spend.
- **ffmpeg-mcp**: MCP server exposing video editing as typed tools callable by LLM agents. TypeScript, FFmpeg, MCP.
- **InkTerm**: themeable terminal with a built-in AI agent and MCP support. TypeScript.
- **RestClient**: desktop REST client with collections, environments, Postman-style test scripts, response timelines and load testing. Tauri, Rust, React, TypeScript.
- **sandkit**: SDK for embedding multi-language runnable code playgrounds, powered by Piston, Pyodide and QuickJS. TypeScript, Hono, Turborepo.
- **dockman** (Tauri, Rust, TypeScript), **redis_gui** (TypeScript, Redis) and **Yankit** (Swift, SwiftUI, macOS): developer tooling for container management, Redis inspection and clipboard workflow.

AWARDS

- **1st runner-up out of 14 teams**, Josys company-wide hackathon, leading a team of 3 against teams mentored by product managers and architects. The project, a dynamic AI-controlled workflow engine on MCP, was **selected for production release ahead of the winning entry**.
- **Top performer rating, Josys, two consecutive years**.

EDUCATION

B.E. Computer Science | East Point College of Engineering and Technology, Bengaluru | 2013 to 2017